

Fig. 5: HDMI orientation

Extended display-identification data

Extended display-identification data (EDID) is a data structure that is stored into the sink device and hence enables a source device to understand the sink device's audio-video capability. EDID data is read by the source device using display data channel lines.

The source device reads EDID from the sink device only when there is a hot-plug detect assertion of minimum 100ms. If there is no such assertion (keeping low hot-plug detect line for minimum 100ms), reading of EDID is not possible. The assertion is triggered by the sink device (like TV) whenever you perform the following actions:

- Removing and inserting HDMI cable
- Switching off and on the TV or source device
- Switching between different HDMI sources
- Soft hot-plug detection that can be done internally by software

You can see the following issues if EDID is not read by the source device or hot-plug detect assertion is not triggered:

- Source device may always send the default resolution to the TV. This can be the only standard-definition resolution
- Source device may not send any audio and video to the TV

- HDCP decryption procedure may fail; hence, you may only see mosquito noise
- TVs and audio systems that support audio return channel (ARC) may not work when EDID is not read
- Source device may not send the audio codec to the TV; hence, only default LPCM audio is sent

So, if you face any of these issues in your TV, you can work-around these by any of the following means:

- Removing and inserting HDMI cable
- Switching off and on the TV or source device
- Switching between different HDMI sources

High-bandwidth digital content protection

High-bandwidth digital content protection (HDCP) is a mechanism to prevent copying of audio and video content travelling from one device to another via HDMI. Hence, any device that does not have high-bandwidth digital content protection support, or if a hacker tries to copy the audio and video content from the source device, audio and video will not be available. HDCP 2.2 is used in HDMI 2.0 and above versions of HDMI.

Electrical/physical protocols

Audio and video is transmitted from the source to sink device by electrical transition minimised differential signaling mechanism. The source device must ensure electrical characteristics during transition minimised differential signaling as shown in Fig. 2.

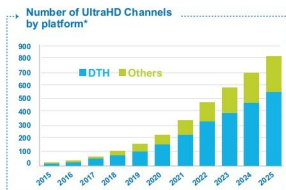


Fig. 6: UHD channel growth

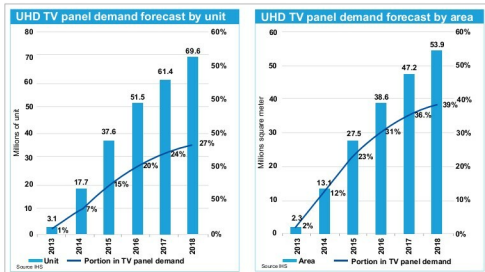


Fig. 7: UHD TV panels demand forecast